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Experimental study and analysis on driving wheels' performance for planetary exploration rovers moving in deformable soil

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Abstract

Planetary rovers are different from conventional terrestrial vehicles in many respects, making it necessary to investigate the terramechanics with a particular focus on them, which is a hot research topic at the budding stage. Predicting the wheel–soil interaction performance from the knowledge of terramechanics is of great importance to the mechanical design/evaluation/optimization, dynamics simulation, soil parameter identification, and control of planetary rovers. In this study, experiments were performed using a single-wheel testbed for wheels with different radii (135 and 157.35 mm), widths (110 and 165 mm), lug heights (0, 5, 10, and 15 mm), numbers of lugs (30, 24, 15, and 8), and lug inclination angles (0°, 5°, 10°, and 20°) under different slip ratios (0, 0.1, 0.2, 0.3, 0.4, 0.6, etc.). The influences of the vertical load (30 N, 80 N, and 150 N), moving velocity (10, 25, 40, and 55 mm/s), and repetitive passing (four times) were also studied. Experimental results shown with figures and tables and are analyzed to evaluate the wheels' driving performance in deformable soil and to draw conclusions. The driving performance of wheels is analyzed using absolute performance indices such as drawbar pull, driving torque, and wheel sinkage and also using relative indices such as the drawbar pull coefficient, tractive efficiency, and entrance angle. The experimental results and conclusions are useful for optimal wheel design and improvement/verification of wheel–soil interaction mechanics model. The analysis methods used in this paper, such as those considering the relationships among the relative indices, can be referred to for analyzing the performance of wheels of other vehicles.
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Keywords: Terramechanics; Planetary exploration rover; Wheel performance; Dimension; Lug; Slip ratio; Performance indices

1. Introduction

Exploring planets such as the Mars and the moon with autonomous wheeled robots (rovers) has already been proved to be an effective method by the successful rovers, including the well-known "Sprit" and "Opportunity" of the US and "Lunakhod" of the former USSR [1,2]. At present, several new rover-based planetary exploration missions are in progress, which require the rovers to travel a much longer distance over more challenging terrain than did earlier missions [3–5]. In order to fulfill complex scientific exploration tasks in rough deformable terrain covered with fine-grained regolith, scientists have been researching and endeavoring to develop high-performance rovers.

Predicting wheel–soil interaction performance from the knowledge of terramechanics is of great importance to mechanical design/evaluation/optimization [6–8], dynamics simulation [9–11], soil parameter identification [12,13], locomotion control [14,15], path following [16], and even to path planning [17] of rovers. Terramechanics theory developed by Bekker et al. [18–22] for conventional terrestrial vehicles was usually directly used for the research on planetary rovers. However, due to the differences between planetary rovers and terrestrial vehicles with regard to the physical dimensions, wheel shape, payload, terrain, running velocity, control mode, and so on, it is necessary to examine the applicability of the conventional theory and improve it so that it can be applied to planetary rovers.

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Nomenclature

$\boldsymbol{W}$	vertical wheel load
$\boldsymbol{f}_{DP}$	horizontal resistance encountered by a wheel
$\boldsymbol{T}$	driving torque generated by the wheel motor and reduction gears
$\boldsymbol{F}_N$	normal force exerted on the wheel by soil
$\boldsymbol{F}_{DP}$	drawbar pull exerted on the wheel by soil
$\boldsymbol{M}_R$	resistance moment exerted on the wheel by soil
k_c	cohesive modulus of soil
k_φ	frictional modulus of soil
n	sinkage exponent of soil
c	cohesion of soil
φ	internal friction angle of soil
K	shearing deformation modulus of soil
r	wheel radius
r_s	equivalent shearing radius of the wheel
λ_s	coefficient used for calculating r_s
b	wheel width
h	lug height
β_L	inclination angle of wheel lugs
n_L	number of wheel lugs
γ_L	spacing angle of wheel lugs, $\gamma_L = 2\pi/n_L$
z	wheel sinkage
θ_1	entrance angle of the wheel moving in soil
θ_2	leaving angle of the wheel moving in soil
θ_m	the angular position of the maximum stress exerted on the wheel by soil
c_1, c_2	coefficients used for calculating θ_m
s	slip ratio
v	linear velocity of the wheel
ω	angular velocity of the wheel
n_p	repetitive times for a wheel travelling on the same terrain
t	time
PC	drawbar pull coefficient
TC	tractive force coefficient
RC	resistance force coefficient
PE	drawbar pull efficiency
TE	tractive efficiency
P	effective power of the motor
L	the distance between the tracks generated by two adjacent lugs

Terramechanics of planetary rovers is a hot research topic at the budding stage, which presents both new challenges and new opportunities to the fields of conventional vehicles and mobile robots [23].

Terramechanics is a subject that closely combines theory and experiments. In order to study the terramechanics of planetary rovers, some testbeds were developed according to the special conditions of planetary rovers [7,24–26] and used to conduct wheel–soil interaction experiments. From the experimental results, the performance of wheels and rovers could be analyzed for understanding the relationships among parameters such as the normal force, drawbar pull, driving torque, wheel sinkage, and slip ratio as well as the influence of wheel parameters (such as the wheel radius/width, and height/number/inclination angle of lugs on wheel performance. The experimental results were also used for deducing, improving, and validating semi-empirical wheel–soil interaction models based on the conventional theory.

Iagnemma et al. simplified conventional stress distribution equations to linear functions according to the observation that stress distributions are approximately linear for most of the natural terrains. By using these simplified equations, they deduced a simplified closed-form formula for wheel–soil interaction mechanics [27]. Further, a linear least-square estimator was adopted to estimate internal cohesion and friction angle on-line by using this formula [28]. The experimental results from a single-wheel testbed [24] confirmed the effectiveness of the simplified equations and the algorithm used for soil parameter estimation. Bauer et al. tested the performance of a wheel with 9 and 18 wheel lugs by using a single-wheel testbed and used the results to validate a dynamic computer simulator [29]. Michaud et al. presented the rover chassis evaluation tools (RCET) and used testbeds to validate the locomotion performance [7,30]. Jizuka et al. developed a pentagon-type wheel for an exploration rover and experimentally studied its slope-climbing performance [31]. Yoshida et al. derived an improved practical model for calculating the drawbar pull as a function of the vertical load and slip ratio on the basis of the Wong–Reece terramechanics formula [32]; in order to analyze the steering performance of a wheel and a rover, the lateral force characteristics of a driving wheel were modeled as a function of the slip ratio and slip angle [33]. The proposed models were validated with reasonable precision by experimental results obtained using a single-wheel testbed [25]. Liu et al. conducted experiments to analyze the effect of slip on the tractive performance of rigid wheels in loose soil [34]. Recommendations for grouser parameters were given: a spacing angle of 15°, grouser height of 10 mm, and grouser thickness of 1.5 mm [35] based on the experiments with a single-wheel testbed [26]. Zou et al. compared the performance of a lugged wheel and a smooth wheel based on experimental results and concluded that the drawbar pull of the former wheel is 5.2 times as that of the latter one [36].

Theoretical prediction of the wheel performance is intrinsically difficult because the wheel–soil interaction process is intricate and is influenced by many factors. The objective of this study is to investigate the effect of several factors on the wheel performance of planetary rovers on the basis of experimental results, including the wheel diameter/width, lug height/number/inclination angle, moving

velocity, slip ratio, and passing times. Absolute performance indices such as the normal force, drawbar pull, driving torque, and wheel sinkage as well as relative indices such as the pull coefficient, tractive efficiency, and entrance angle are utilized to evaluate the performance of wheels. The experimental results and conclusions could be applied in analyzing the wheel–soil interaction process, optimizing wheel design, and improving/verifying a wheel–soil interaction mechanics model.

2. Influential factors and evaluation indices for driving wheel's performance

Fig. 1 shows the forces and torque exerted on a driving wheel by the soil, vehicle body, and driving motor.

For a wheel moving with a constant velocity, the following equilibrium equations can be obtained:

$$\begin{cases} F_N = F_N(P_S, P_W, P_L, P_R) = W \\ F_{DP} = F_{DP}(P_S, P_W, P_L, P_R) = f_{DP} \\ M_R = M_R(P_S, P_W, P_L, P_R) = F_{DP} \cdot l + F_N \cdot e = T \end{cases} \quad (1)$$

where $P_S = \{k_c, k_\varphi, n, c, \varphi, K, c_1, c_2\}$ and represents the soil parameters, $P_W = \{r, b\}$ and represents the wheel dimensions, $P_L = \{h, \beta_L, \gamma_L(n_L)\}$ and denotes the wheel lug parameters, and $P_R = \{\theta_1, \theta_2, s, v, n_p\}$, including the running state variables. F_N, F_{DP}, and M_R denote the equivalent forces and moment exerted on the wheel center by the soil.

In fact, the soil exerts the continuous normal stress $\sigma(\theta)$ and the shearing stress $\tau(\theta)$ on the wheel; the concentrated forces and moment are the integration of distributed stresses [27]. The entrance angle θ_1 can be calculated with the following equation:

$$\theta_1 = \arccos(1 - z/r) \quad (2)$$

The maximum stress angle θ_m is usually considered as a linear function of θ_1:

$$\theta_m = (c_1 + c_2 s)\theta_1 \quad (3)$$

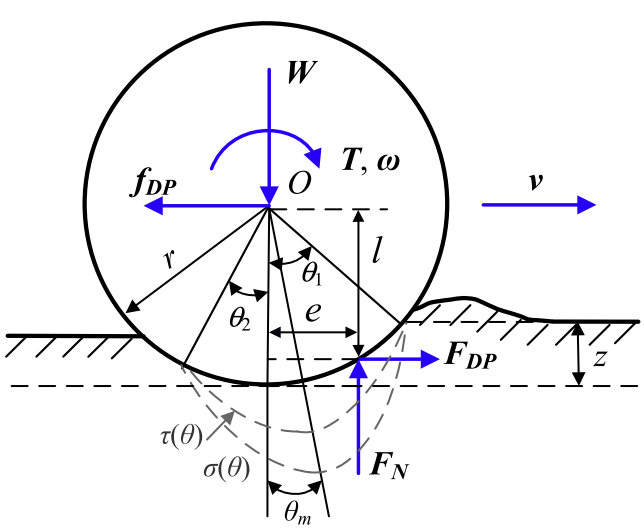


Fig. 1. Forces and torque acting on a driving wheel.

The leaving angle θ_2, which is usually simplified to zero because it is small and not easy to determine, is the angle of the rear part where stresses are zero. θ_2 can also be calculated by multiplying θ_1 with a proportionality factor [37].

The slip ratio s is an important state variable for wheel–soil interaction, which is defined as follows [38]:

$$r_s = (r + \lambda_s h)(0 \leqslant \lambda_s \leqslant 1) \quad (4)$$

$$s = \begin{cases} (r_s\omega - v)/r_s\omega & (r_s\omega \geqslant v, 0 \leqslant s \leqslant 1) \\ (r_s\omega - v)/v & (r_s\omega < v, -1 \leqslant s < 0) \end{cases} \quad (5)$$

The shearing radius r_s is an equivalent radius where shear occurs between the moving soil and the static soil. It can be determined by assuming the drawbar pull to be zero for zero slip. The value of λ_s is an increasing function of the number of lugs and the internal friction angle of the soil. If the number of lugs is sufficient and the internal friction angle is large, causing most of the soil between two adjacent lugs to move with the wheel, s_3 is near to 1.

The mechanical property parameters of planetary soil can be considered as known despite the fact that the soil parameters may be different to some extent from place to place. In order to optimize the wheel design so as to satisfy the demands of challenging exploration missions, it is important to understand the effects of the dimension parameters P_W and the wheel lug parameters P_L on the wheel performance. Experiments for wheels with different parameters of r, b, h, β_L, and n_L should be carried out to study their effect on the wheel performance.

For a certain wheel travelling on the planetary soil, the parameters of P_S, P_W, and P_L can be regarded as known. Then, the wheel–soil interaction forces and moment are primarily determined by the running state variables. As T is the function of F_N and F_{DP}, there are only two independent equations in Eq. (1). For an exploration rover, the forces of F_N and F_{DP} can be considered as independent variables, which are mainly determined by the payload, terrain geometry, and rover configuration. The angular velocity of a wheel, to which the linear velocity v is proportional, is set by the operators. If the moving velocity v and passing times n_p are known, the three remaining unknown variables, i.e., the driving torque T and state variables s and θ_1, are mutually dependent and can be solved with Eq. (1). While carrying out experiments, the slip ratio s is set up as an independent variable instead of F_{DP} by modulating the angular velocity of a driving motor and a carriage motor. The normal force F_N (equal to the vertical load W on flat terrain) is set up by adjusting the balance weight.

If the vertical load W and the slip ratio s are given, the drawbar pull F_{DP} exerted on the wheel by the soil, the required driving torque T (equal to the resistance torque M_R), and the wheel sinkage z (or the entrance angle θ_1) can be directly measured to evaluate the performance of the wheel. In order to compare the performance of wheels with different design parameters or payloads, some relative indices should be adopted. The performance evaluation indices of a driving wheel are summarized as follows.

2.1. Indices of wheel sinkage

While moving on the deformable soil of rough terrain, the rover wheels may suffer severe slip-sinkage and, in the worst case, even the rover may get stuck in the soil. Slip-sinkage is an important failure form for planetary rovers. For example, the "Opportunity" Mars rover required about 5 weeks to escape from the Purgatory Dune after getting stuck in 2005. The wheel sinkage z is usually utilized to describe the degree of sinkage. It was noticed that the wheel lugs have little influence on the sinkage by sustaining the wheel [39]. Hence, the wheel sinkage z should not include the height of lugs. For wheels with different radii, the same wheel sinkage can affect the wheels differently. The parameter z/r describes the degree of relative sinkage. The entrance angle θ_1, which is a trigonometric function of z/r, as shown by Eq. (2), is adopted in this paper, for it is a more intuitive index to characterize the relative sinkage.

2.2. Indices of drawbar pull performance

As the value of e/r_s is small, l is approximately equal to r_s. By replacing l with r_s in Eq. (1), one obtains

$$T = F_{DP} \cdot l + F_N \cdot e \approx F_{DP} \cdot r_s + F_N \cdot e \tag{6}$$

$$F_{DP} \approx T/r_s - F_N \cdot e/r_s \tag{7}$$

Let $F_t = T/r_s$, denoting the traction force caused by the soil deformation while the motor drives the wheel to rotate. $F_N \cdot e/r_s$ is the resistance caused by the soil, where e/r_s is the resistance coefficient denoted by RC, reflecting the hindering effect of the soil on the wheel. The drawbar pull F_{DP} is the net force for driving the rover forward.

Divide Eq. (7) by F_t, one obtains

$$F_{DP}/F_t + F_N \cdot RC/F_t \approx 1 \tag{8}$$

Let $PE = F_{DP}/F_t = F_{DP}r_s/T$, denoting the drawbar pull efficiency. PE is the efficiency with which the tractive force is transformed into the effective drawbar pull.

Divide Eq. (7) by F_N, one obtains

$$F_{DP}/F_N \approx T/(F_N r_s) - RC \tag{9}$$

Let $PC = F_{DP}/F_N$, denoting the drawbar pull coefficient. PC is the drawbar pull caused by a unit normal force. Let $TC = T/(F_N r_s)$, denoting the tractive coefficient, implying the tractive force caused by the torque T of the wheel's motor and reduction gears for a unit normal force. According to the above analysis,

$$PC \approx TC - RC \tag{10}$$

$$PE = \frac{F_{DP}/F_N}{T/F_N r_s} = PC/TC \tag{11}$$

Not only that PC can be used to compare the pulling ability of a wheel under different vertical loads, but also it is an index for evaluating the trafficability such as the slope-climbing ability of a rover over rough terrain. Fig. 2 shows the equivalent forces and moment acting on a wheel while climbing up the slope. The relationship between PC and α is

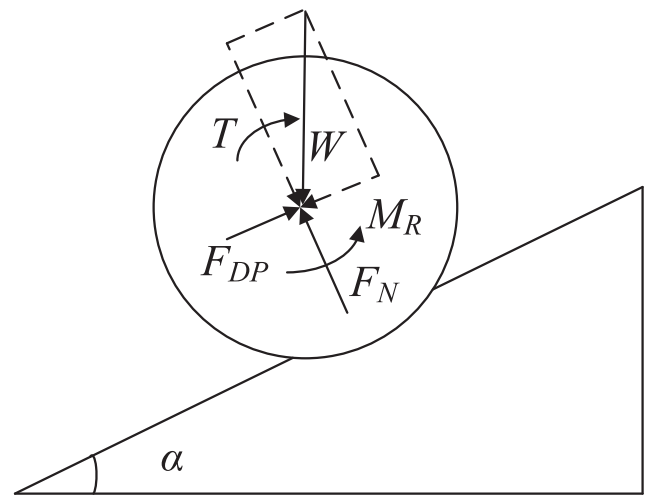


Fig. 2. Force diagram for a wheel climbing up a slope.

$$PC = \frac{F_{DP}}{F_N \cos\alpha} = \frac{W \sin\alpha}{W \cos\alpha} = \tan\alpha \tag{12}$$

In order to climb up the slope with an inclination angle of α, the maximum PC that a wheel can obtain should exceed $\tan\alpha$. That is, the maximum slope that a rover can climb up is

$$\alpha_{\max} = \arctan(PC_{\max}) \tag{13}$$

The relative error caused by the approximation of Eq. (6) is

$$Err = \frac{F_{DP}(r_s - l)}{F_{DP} \cdot l + W \cdot e} = \frac{r_s - \sqrt{r_s^2 - e^2}}{\sqrt{r_s^2 - e^2} + e/PC} = \frac{1 - \sqrt{1 - RC^2}}{\sqrt{1 - RC^2} + RC/PC} \tag{14}$$

Err denotes an increasing function of RC and PC. Fig. 3 shows the plots of Err versus the slip ratio for three wheels with the data obtained from experiments described in the

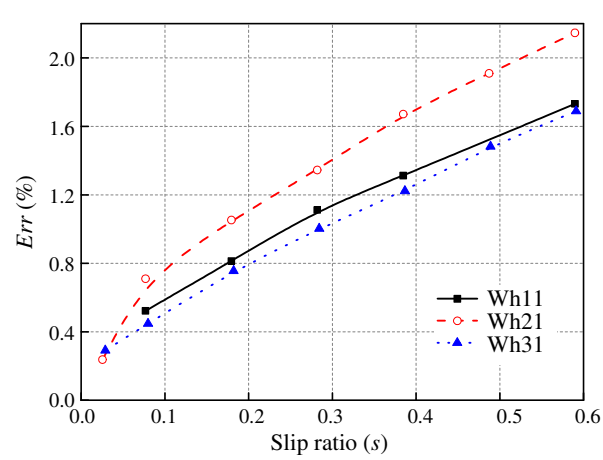


Fig. 3. Relative approximation error of Eq. (6).

following sections. If the slip ratio is 0.6, the relative errors are maximum with values of approximately 2%. Therefore, the simplification error is negligible.

The indices F_{DP}, PC, RC, and PE can be used to evaluate the drawbar pull performance of a wheel.

2.3. Motor performance indices

The motor performance indices are related to the motor of the wheel. The driving torque T and the effective power P are two important absolute indices. The tractive force coefficient TC and the tractive efficiency TE, which is used to evaluate the efficiency of the driving motor, are relative indices. Equations for calculating P and TE are as follows:

$$P = T\omega = Tv/[r_s(1-s)] = TC \times F_N v/(1-s) \tag{15}$$

$$TE = \frac{F_{DP} \cdot v}{T\omega} = \frac{F_{DP} \cdot r_s}{T}(1-s) = PE(1-s) \tag{16}$$

According to the above analysis, if the slip ratio is known, only two of the five relative performance indices, i.e., PC, TC, RC, PE, and TE, are independent. The other indices can be calculated with Eqs. (10), (11), and (16).

3. Experimental setup

3.1. Wheel–soil interaction testbed

The wheel–soil interaction test system developed in Chinese State Key Laboratory of Robotics and System of Harbin Institute of Technology (as shown in Fig. 4) [26] was used to measure the wheel sinkage and related forces/torques. The testbed has dimensions of 1700 mm × 850 mm × 900 mm. It can be used to perform experiments on wheels with a radius ranging from 50 mm to 200 mm and a width ranging from 60 mm to 300 mm. The testbed contains three motors (driving motor, steering motor, and carriage motor), instrumented with a displacement sensor, six-axis F/T sensor, driving torque sensor, current sensors, and optical encoders. The driving motor can cause the wheel to move forward. The carriage motor is used together with a conveyance belt that imitates the influence of the vehicle body on the wheel in order to create various slip ratios. The steering motor can drive the motor to steer. The wheel sinkage is measured by a high-precision linear-potentiometer displacement sensor. An F/T sensor can measure the drawbar pull, steering torque, side force, normal force, etc., while the driving torque of a motor is measured by a torque sensor. The current sensors and optimal encoders are used to measure the current and position of each motor.

Fig. 4. Wheel–soil interaction testbed.

3.2. Experimental wheel and planetary soil simulant

Three types of cylindrical metal wheels with different dimensions ($r \times b$) are used: 135 mm × 165 mm (Wh1), 135 mm × 110 mm (Wh2), and 157.35 mm × 165 mm (Wh3). Lugs differing in heights (5, 10, and 15 mm) and numbers (8, 15, 24, 30, etc.) could be installed on these smooth wheels. The wheel dimensions are comparable to those of recent planetary rovers. The wheels and lugs are combined according to the research purpose of the experiments, as tabulated in Table 1. Fig. 5 shows two of the experimental wheels.

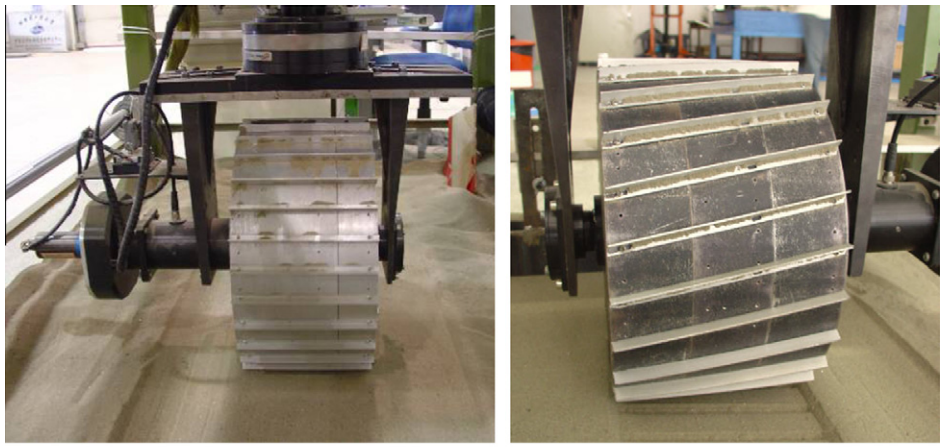
The planetary soil simulant is made from soft sand after removing impurities and performing sieving, ventilating, and drying; the sand is provided by Shanghai Academy of Spaceflight Technology. The physical and mechanical property parameters, which are measured by plate-sinkage experiments, shearing experiments, and a parameter identification method, are listed in Table 2. The mechanical and physical properties are close to those of the lunar soil.

3.3. Experimental methodology

It is essential to obtain data for analyzing the effects of wheel parameters and running state variables on wheel performance. However, investigation of all combinations of these parameters and variables will require several thou-

Table 1
Wheel codes and parameters.

Wheel codes	r (mm)	b (mm)	h (mm)	n_L	β_L (°)
Wh11	**135**	165	**15**	24	**0**
Wh112	135	165	15	24	**5**
Wh113	135	165	15	24	**10**
Wh114	135	165	15	24	**20**
Wh12	**135**	165	**10**	24	0
Wh13	135	165	**5**	24	0
Wh14	135	165	**0**	–	–
Wh21	135	**110**	15	24	0
Wh22	135	**110**	10	24	0
Wh31	**157.35**	**165**	**15**	**30**	0
Wh312	157.35	165	15	**24**	0
Wh313	157.35	165	15	**15**	0
Wh314	157.35	165	15	**8**	0
Wh32	**157.35**	**165**	**10**	30	0
Wh33	157.35	165	**5**	30	0
Wh34	157.35	165	**0**	–	–



(a) Wh32 (b) Wh114

Fig. 5. Experimental wheels.

Table 2
Physical and mechanical parameters of planetary soil simulant [42,43].

ρ (kg/m^3)	k_c (kPa/m^{n-1})	k_φ (kPa/m^n)	n	c (kPa)	φ (°)	K (mm)
1.605	15.6	2407.4	1.10	0.25	31.9	9.7–13.1

sands of experiments. In such a situation, experimental design methods such as the orthogonal experimental method seem effective in decreasing the experimental combinations. But all the concerned parameters are quite different in essence, and this study will not only investigate the influence of each parameter but also provide a series of effective experimental results for the development of mathematical models. In order to obtain useful experimental results by conducting a limited number of experiments, the experiments were designed based on theoretical analysis and were optimized according to the conclusions drawn from the results of the preceding experiments.

The velocity of current planetary rovers is very low. The MER rovers have a top speed on flat hard ground of 50 mm/s [1], and the lunar rover under development in China has a comparable top velocity. In order to test the velocity effect, the travelling velocity of a wheel was set to be 10, 25, 40, and 55 mm/s.

During the driving process of a wheel, the slip ratio is usually not smaller than zero. While moving, the slip ratio of a rover should be controlled to be less than a certain value, as slip-sinkage is quite serious for a rigid wheel moving in deformable soil with a high slip ratio. After the Spirit rover got stuck in the Mars sand, NASA measures the slip ratio of the MER rovers by visual odometry so as to restrict it to a value smaller than 0.4 while exploring [40–41]. Accordingly, in this study, the experimental slip ratios were set to 0, (0.05), 0.1, 0.2, 0.3, 0.4, (0.5), 0.6, (0.7), and (0.8). In addition, some negative slip ratios (i.e., skid) were set to test the influence of velocity. In order to compare the performance of wheels with different lug heights, the slip ratios were amended according to the experimental results by using the equivalent shearing radii for drawing figures.

In order to test the effect of repetitive passing, the wheel is controlled to traverse the same path for four times under different slip ratios.

The effects of the wheel width, wheel diameter, lug height, lug number, and lug inclination angle are investigated by using a wheel with different dimension parameters and lug parameters.

The vertical load is set to be 80 N according to the ability of the testbed, which is comparable to that of planetary rovers. In order to investigate the effect of the vertical load on wheel performance, the vertical loads of 30 N and 150 N were also adopted for the wheel Wh32.

The experimental description, group number, test number, velocity, wheel code, vertical load, and slip ratio are tabulated in Table 3.

3.4. Experimental procedure

The experimental procedure is described as follows. (1) Select a wheel and install on it a certain number of lugs having a certain height. (2) Adjust the weights of the counterweight mechanism so as to obtain a vertical load of 30 N, 80 N, or 150 N. (3) Set the parameters such as the wheel radius, lug height, moving velocity, and slip ratio with the control software. (4) Make the soil uniform and smooth with the strickling mechanism. This is very important to ensure the reproducibility of the experimental results. (5) Calibrate the sensors. While the wheel was suspended, the reference values of the F/T sensor and torque sensor can be obtained easily by clicking the "Calibration" button of the control software. (5) Lower the wheel so that it makes contact with the soil by adjusting the height of the sustaining jack. (6) Start testing, import data in real-time,

Table 3
Design of experiments.

Description	Group no.	Test no.	v (mm/s)	Wheel code	W (N)	s
Influence of velocity	G1	T1–T4	**10**, **25**, **40**, **55**	Wh12	80	−0.4, −0.2, 0, 0.2, 0.4
	G2	T5–T8	**10**, **25**, **40**, **55**	Wh14	80	
Repetitive passing effect	G3	T9–T12	25	Wh12	80	0.1, 0.2, 0.3, 0.4
Effect of lug height	G4	T13	10	**Wh34**	80	0, (0.05), 0.10, 0.20, 0.30, 0.40, (0.50), 0.60 (0.70), (0.80)
		T14		**Wh33**		
		T15		**Wh32**		
		T16		**Wh31**		
Effect of lug number	G5	T17	10	**Wh31**	80	
		T18		**Wh312**		
		T19		**Wh313**		
		T20		**Wh314**		
Effect of lug inclination angle	G6	T21	10	**Wh114**	80	
		T22		**Wh113**		
		T23		**Wh112**		
		T24		**Wh11**		
Influence of wheel diameter	G7	T25	10	**Wh12**	80	
		T15		**Wh32**		
	G8	T26	10	**Wh11**	80	
		T16		**Wh31**		
Influence of wheel width	G9	T27	10	**Wh22**	80	
		T25		**Wh12**		
	G10	T28	10	**Wh21**	80	
		T26		**Wh11**		
Influence of vertical load	G11	T29	10	Wh32	**30**	
		T15			**80**	
		T30			**150**	

and stop the motor in the end. (7) Adjust the height of the jack to make the wheel suspended and return it back to the initial position. (8) Repeat the above steps to finish all the experiments.

3.5. Experimental results

The experimental data were acquired by a commercial data acquisition card. The data sampling frequency was set to 6.67 Hz, so that hundreds of original data values could be obtained for a test. In order to ensure the repeatability of the experiments, many of the tests were carried out twice or thrice. It was verified that the experiments have good repeatability.

While moving, the wheel lugs enter and exit the soil periodically and leave traces (generate tracks) on the soil. Fig. 6 shows the traces of Wh32 under the slip ratios of 0.1 and 0.4. The track generated by the lugs becomes denser with an increase in the slip ratio. L, the distance between the tracks generated by two adjacent lugs can be calculated with Eq. (17):

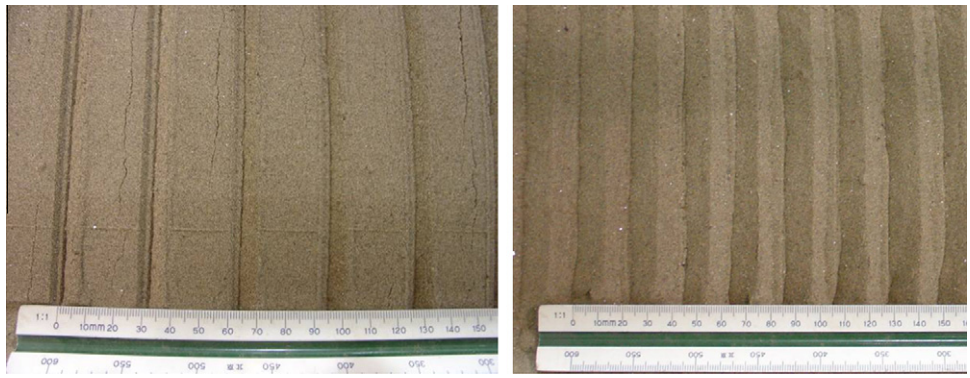
$$L = \begin{cases} (1-s)r_s\gamma & (0 \leqslant s \leqslant 1) \\ r_s\gamma/(1+s) & (-1 \leqslant s < 0) \end{cases} \tag{17}$$

The experimental and theoretical results of L agree very well, which can be used in turn to estimate the slip ratio of a wheel [38].

Because of the periodical contact of the wheel lugs with the soil, the measured data such as the wheel sinkage, drawbar pull, and driving torque fluctuate in the form of sinusoids. The period of the fluctuation is

$$T = \frac{\gamma}{\omega} = \begin{cases} r_s\gamma(1-s)/v & (0 \leqslant s \leqslant 1) \\ r_s\gamma/[v(1+s)] & (-1 \leqslant s < 0) \end{cases} \tag{18}$$

Although the measured data fluctuate with large amplitudes, the filtered results are relatively smooth, as shown in Fig. 7. A Matlab GUI program was developed to draw the curves of the measured data versus time, calculate the mean value and standard deviation of the data that fluctuate steadily (for example, the data obtained from 10 to 40 s in Fig. 7). The wheel sinkage, drawbar pull, and driving torque corresponding to various wheels, lugs, slip ratios, and weights were obtained after processing the data acquired in all the experiments. Fig. 8 shows an example of the processed data on drawbar pull and driving torque for Wh32 with a vertical load of 80 N. The mean values of hundreds of raw data values are plotted in the following figures, while the error bars are not shown for the sake of clarity.



(a) s=0.1 (b) s=0.4

Fig. 6. Traces of Wh32 (W = 80 N).

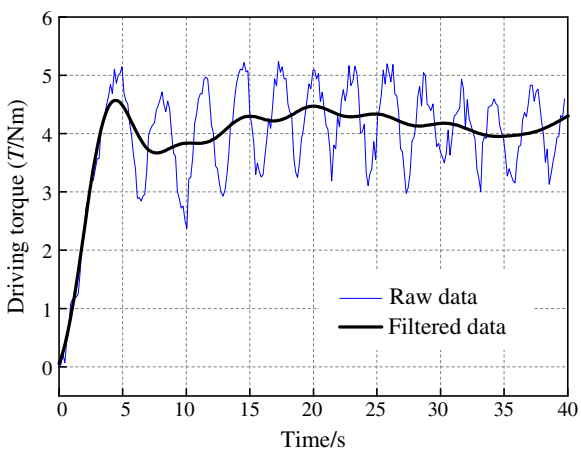


Fig. 7. Raw data and filtered data of torque for wheel Wh12 (F_N = 80 N, s = 0.2).

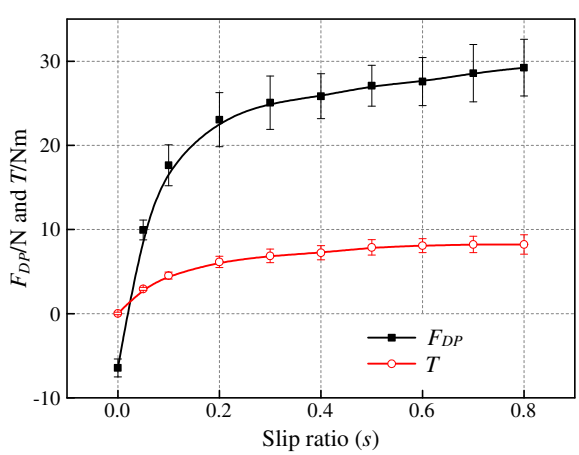


Fig. 8. F_{DP} and T with error bar for Wh32 (F_N = 80 N).

4. Performance analysis of wheels with different lugs

4.1. Effect of lug height

Fig. 9 shows the curves of F_{DP}, T, and z versus the slip ratio for wheels with lugs of different heights (Wh3). The driving torque obviously increases with the lug height; the drawbar pull also increases as a result. The maximum drawbar pull of the wheel with lugs of 15-mm height is increased by 60% as compared with that of the corresponding smooth wheel. The drawbar pull coefficient PC increases with the same trend as does F_{DP} because the normal force is almost the same. The drawbar pull coefficient PE and tractive coefficient TE were calculated and it was found out that the lug height has relatively small influence on them. Fig. 15 shows the PE and TE for wheels with lugs of 10-mm and 15-mm height.

Fig. 10 shows the influence of the lug height on the raw data on the drawbar pull and driving torque. The wheel sinkage fluctuates periodically in accordance with the entrance and exit of the wheel lugs. The resulting dynamic load causes the fluctuation in the normal force, drawbar pull, and driving torque exerted on the wheel by the soil. As observed from the standard deviation (SD) and relative standard deviation (RSD) of the measured data in Table 4, the fluctuation amplitudes increase with the lug height; this acts against the stabilization of the rover vehicle and payloads. The values of RSD can be fitted with the linear function of $\lambda_h = h/r$:

$$RSD = a_{RSD} + b_{RSD} \times \lambda_h \tag{19}$$

The parameter values of a_{RSD} and b_{RSD} for F_N, z, F_{DP}, and T are 6.69, 0.97; 0.63, 0.31; 9.68, 0.42; and 6.85, 0.53, respectively. As the fluctuation amplitude is proportional to λ_h, the lug height should be restricted to decrease the fluctuation in the mechanics and wheel sinkage during wheel–soil interaction.

4.2. Effect of lug number

Fig. 11 shows the curves of F_{DP}, T, and z versus the slip ratio for wheels with lugs of different number (Wh3, 80 N). The number of lugs has little influence on wheel sinkage. F_{DP} and T that acted on the wheel with 24 lugs were near to those that acted on the wheel with 30 lugs. If a wheel was installed with 8 lugs, the measured F_{DP} and M_R were much smaller. This implies that the shearing between the wheel

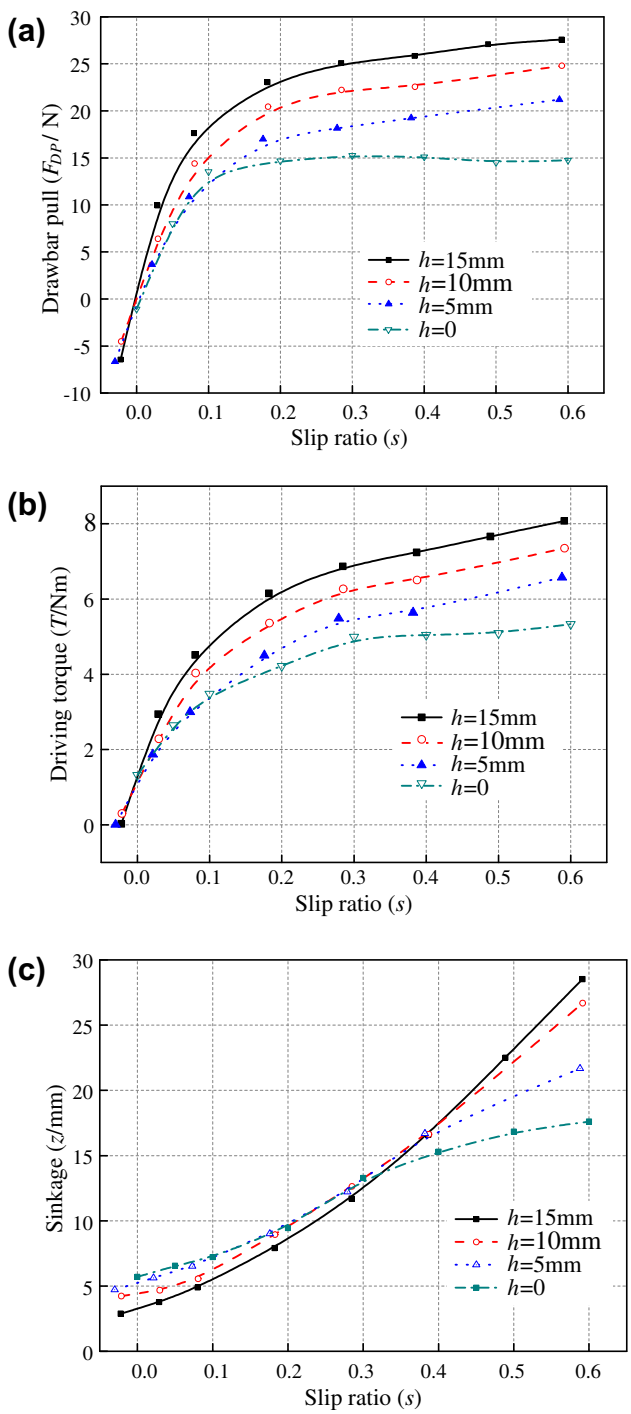


Fig. 9. Curves of F_{DP}, T and z versus slip ratio for wheel Wh3 with lugs of different heights ($F_N = 80$ N).

and the soil is steady and enables to achieve the maximum tractive performance from the soil if the number of wheel lugs exceeds 24. Increasing the number of wheel lugs is beneficial to develop continuous shearing between the wheel lugs and the soil so as to improve the drawbar pull and the coefficient PC until they reach a certain critical value. However, the effect achieved by increasing the lug number is not as much as that achieved by increasing the lug height.

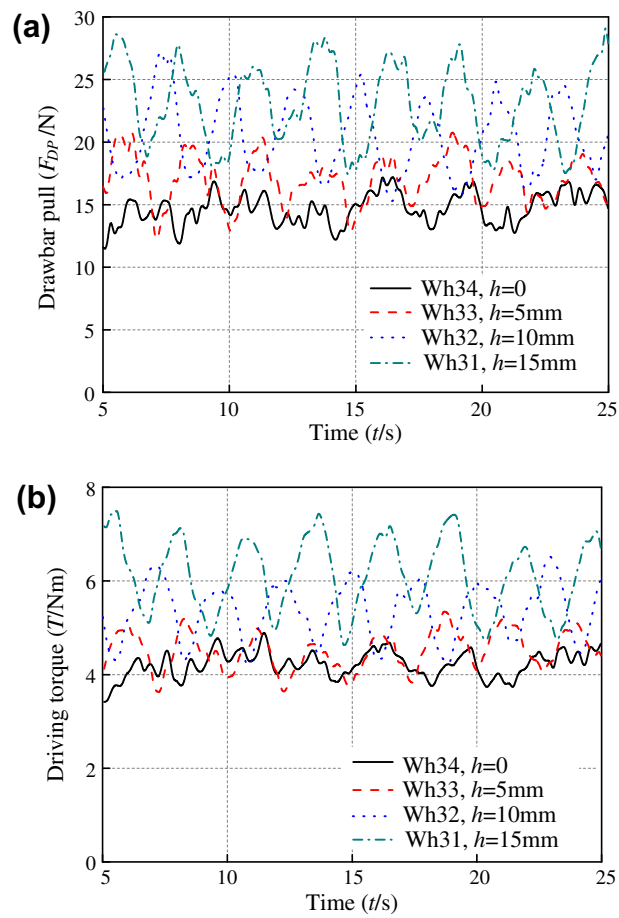


Fig. 10. Influence of lug height on raw data of F_{DP} and T ($s = 0.2$, $F_N = 80$ N).

According to Eq. (18), the fluctuation frequency of raw data increases with the of lug number.

While the slip ratio is between 0.1 and 0.2, the lug number influences F_{DP} and T most obviously. With an increment in the slip ratio, its influence decreases. If the slip ratio attains a value of 0.6, its influence on F_{DP} and T almost disappears. Fig. 12 shows the original data on wheel sinkage and drawbar pull for the wheel with 8 lugs while the slip ratios are 0.2, 0.4, and 0.6. If the slip ratio is very small, the time span of wheel lugs in the soil, which is implied by the smaller wheel sinkage, is very small, but the time span of the interaction between the smooth wheel surface and the soil is very long. With an increment in the slip ratio, the wheel sinkage increases, and the time span of the interaction between the smooth wheel surface and the soil decreases. If the slip ratio is 0.6, at least one lug stays in the soil at any time and the influence of the wheel lug number on wheel–soil interaction is quite small, as shown in Figs. 11 and 12.

4.3. Effect of lug inclination angle

Fig. 13 shows the raw data curves of F_{DP} and T versus the sampling time for wheels with the lug inclination angles

Table 4
Influence of lug height on the *SD* and *RSD* for F_N, z, F_{DP} and T.

h (mm)	λ_h (%)	Standard error				Relative standard error (%)			
		F_N (N)	z (mm)	F_{DP} (N)	T (N m)	F_N	z	F_{DP}	T
0	0	5.15	0.05	1.41	0.30	6.35	0.56	8.96	6.71
5	3.18	7.95	0.16	2.06	0.43	10.16	1.56	11.85	8.83
10	6.35	10.71	0.25	2.93	0.63	13.11	2.89	12.81	10.02
15	9.53	12.83	0.29	3.36	0.81	15.64	3.35	13.05	11.89

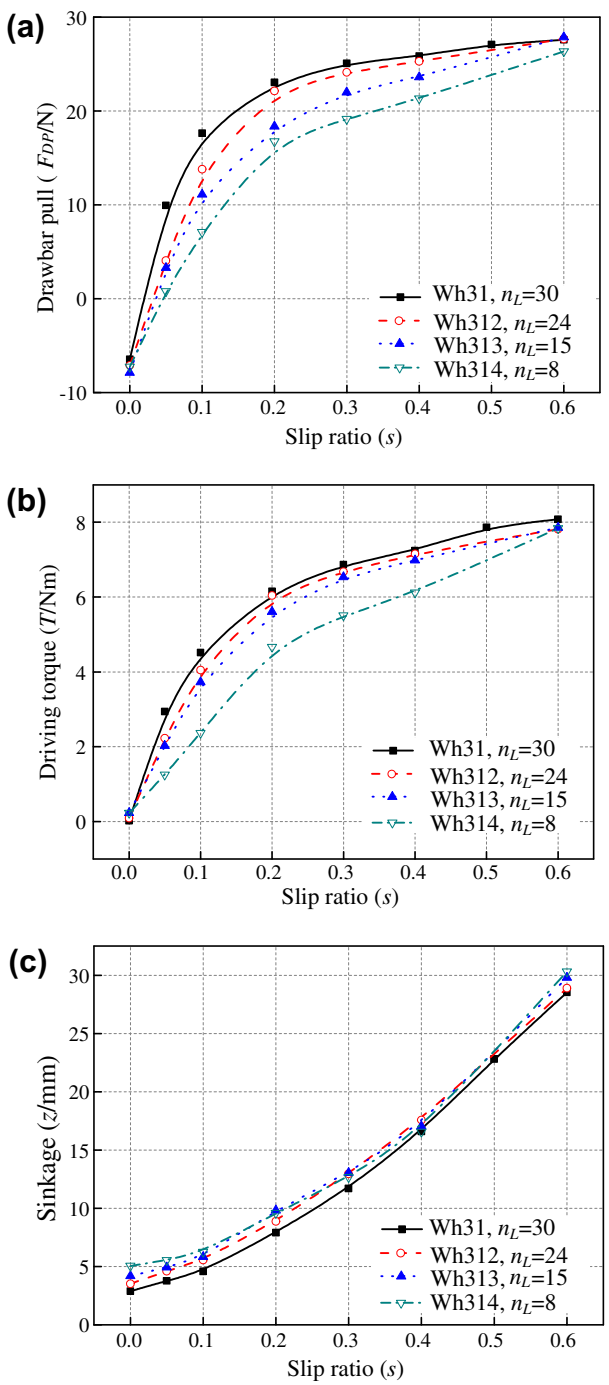


Fig. 11. Curves of F_{DP}, T and z versus slip ratio for wheels with lugs of different number ($F_N = 80$ N).

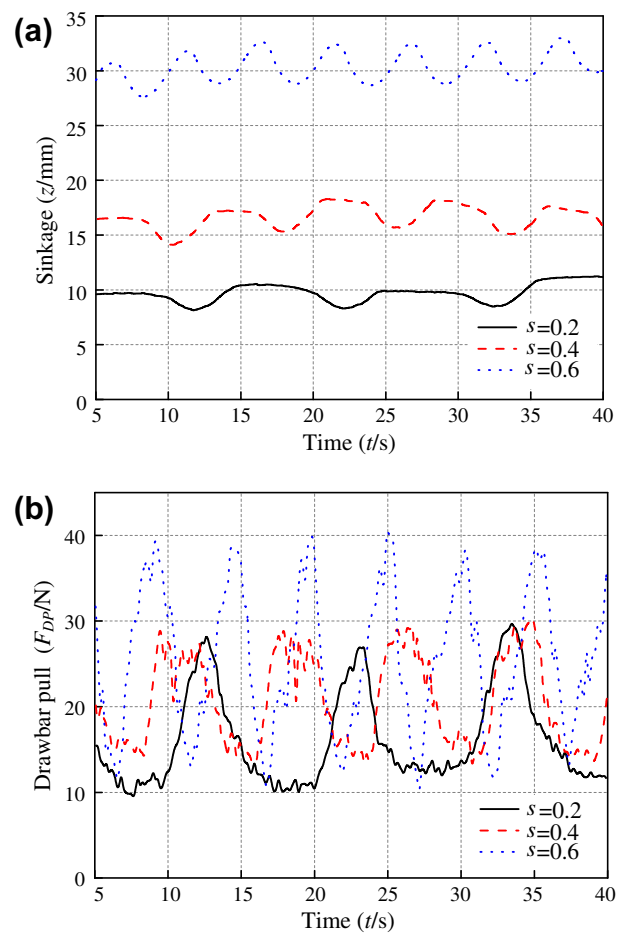


Fig. 12. Raw data of z and F_{DP} and for Wheel Wh314 under different slip ratios ($F_N = 80$ N).

β_L of 0°, 5°, 10°, and 20° ($s = 0.2$). Table 5 lists the influence of the lug inclination angle on the average value and standard deviation of the measured data. The variance of β_L has random influence on the average values, while the values of standard deviation are increasing functions of the angle β_L. For wheels installed with lugs having inclination angles of 10° and 20°, the standard deviations are comparable, which they are much smaller than those for wheels with β_L of 5° and 0°. If β_L exceeds 10°, the shearing between the lugs and the soil is continuous with little sudden impact similar to the engagement of helical gears [44]. However, the inclinations of wheel lugs may generate

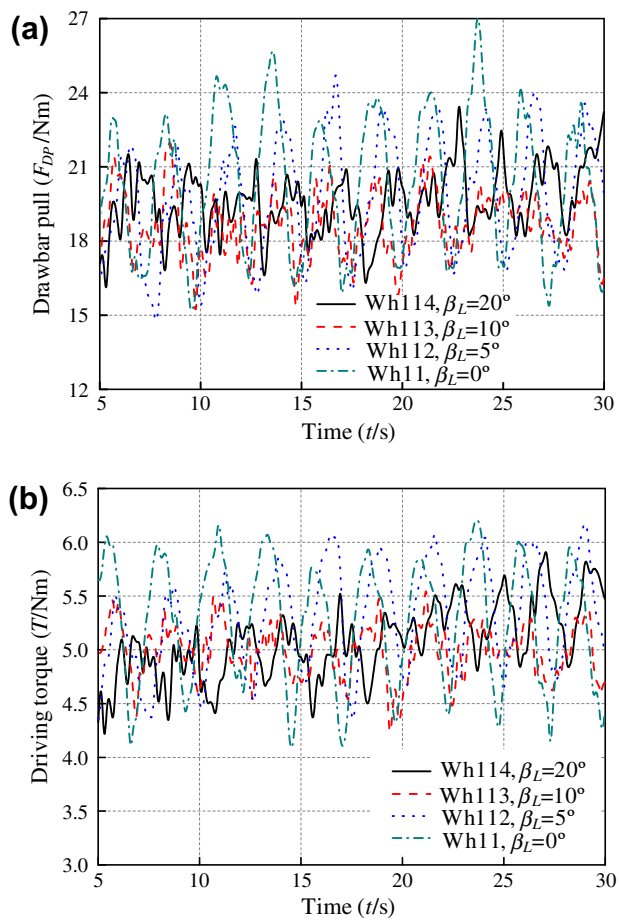


Fig. 13. Influence of lug inclination angle on raw data of F_{DP} and T ($s = 0.2$).

lateral force and cause the mechanical performance of the wheel to be different while the wheel is moving forward and backward. However, if a wheel has a sufficient number of lugs to cause continuous shearing, the wheel and the soil between the lugs can be considered as a cylindrical wheel, so that the influence of lug inclination is small and negligible.

5. Driving performance of wheels with different dimensions

5.1. Effect of wheel radius

Experimental data obtained from the wheels Wh1 and Wh3, which have different radii, are used to analyze the effect of the wheel radius on the driving performance. The radius of Wh1 is 135 mm, and that of Wh3 is 157.35 mm, increased by 16.6%. The width of both the wheel is 165 mm. The wheels were installed with lugs having heights of 15 mm (Wh11, Wh31) and 10 mm (Wh12, Wh32). The absolute and relative performance indices of the wheels are shown in Figs. 14 and 15, respectively.

As observed from Fig. 14, if the slip ratio is 0.6, for the wheels Wh11 and Wh31, the wheel sinkage decreases from 30.6 mm to 28.5 mm (6.9%), the drawbar pull increases from 23.5 N to 27.6 N (17.4%), and the driving torque increases from 6.5 Nm to 8.1 Nm (24.6%); while for wheels Wh12 and Wh32, the wheel sinkage decreases from 28.1 mm to 24.8 mm (11.7%), the drawbar pull increases from 22.6 N to 24.8 N (9.7%), and the driving torque increases from 6.1 Nm to 7.4 Nm (21.3%).

It is seen from Fig. 15a that the entrance angles of different wheels are approximately linear functions of the slip ratio. A wheel with a larger radius has a smaller entrance angle with a difference of about 4° under the same slip ratio, while the slopes of the entrance angle versus the slip ratio plots are almost the same. Fig. 16 shows that the resistance coefficient RC is an increasing function of the entrance angle, which is not influenced so much by the wheel radius. As an increase in the radius can lead to a decrease in the wheel's entrance angle, RC of a wheel with a larger radius is smaller if the slip ratio is same, as shown in Fig. 15b. RC is approximately a linear function of the slip ratio, the slope of which is little influenced by the wheel radius. Fig. 15c shows that the tractive coefficient TC is also seldom influenced by wheel radius, so that an increase in the wheel radius influences the drawbar pull coefficient PC in contrast to the result obtained for RC according to Eq. (11). According to Eq. (15), the dimension effect on the effective power is also very small as F_N is almost constant. As shown in Fig. 15d, an increase in the wheel radius can improve the pull coefficient of a wheel. When the slip ratio is 0.6, PC increases from 0.3 to 0.32 ($h = 15$ mm) and from 0.27 to 0.29 ($h = 10$ mm). However, PC of the wheel Wh11 ($r = 135$ mm, $h = 15$ mm) is even larger than that of the wheel Wh32 ($r = 157.4$ mm, $h = 10$ mm), which indicates the possibility of improving wheel performance by increasing the lug height, and also proves that the effect of increasing the lug height is more than that of increasing the wheel radius. According to $PE = PC/TC$, the influence of radius on PE is similar to that on PC. PE attains the maximum value with a slip ratio between 0.05 and 0.2

Table 5
Influence of lug inclination angle on the SD of F_N, z, F_{DP} and T.

β_L (°)	Average value				Standard deviation			
	F_N (N)	z (mm)	F_{DP} (N)	T (N m)	F_N (N)	z (mm)	F_{DP} (N)	T (N m)
0	81.71	14.90	20.35	5.27	6.311	0.204	2.931	0.599
5	81.91	16.20	19.74	5.30	5.835	0.168	2.478	0.498
10	81.81	15.56	18.66	4.99	3.937	0.093	1.600	0.358
20	77.01	13.47	18.85	5.08	3.791	0.069	1.561	0.345

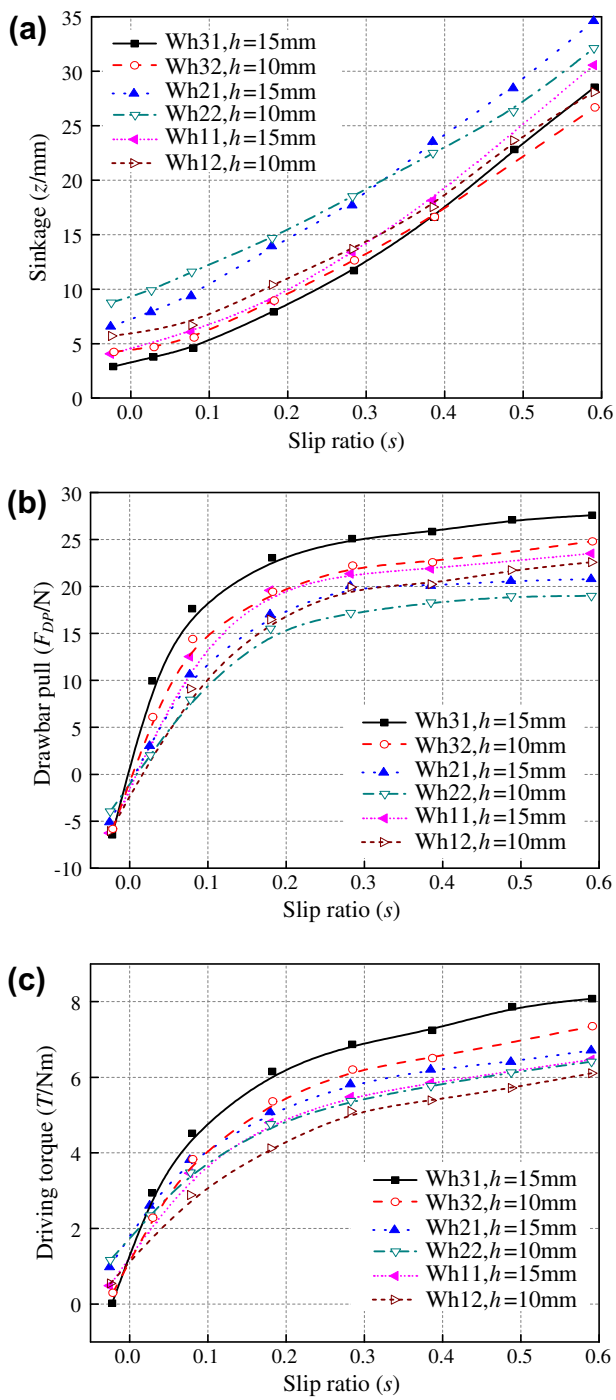


Fig. 14. z, F_{DP} and T for wheels with different dimensions.

and then decreases slightly. The tractive efficient TE also attains the maximum value if the slip ratio is in the range of 0.05–0.2, but then, it decreases rapidly with an increase in the slip ratio.

An increase in the wheel radius can lead to an increase in the length of resistance moment arm, so that a larger driving torque is required to generate the same drawbar pull. If the driving torques are same, the wheel with a smaller radius could obtain larger PC and it can climb up a slope with a larger angle. The angles of slope that a wheel could climb up were calculated by substituting PC into Eq. (13). Fig. 17 shows the curves of the slope-climbing angle versus the driving torque. When the driving torque is 6 N m, the wheel Wh31 could climb up a 14.7° slope, while Wh11 could climb up a 15.5° slope. The corresponding slope-climbing angles of Wh32 and Wh12 are 14.5° and 15.2°, respectively.

5.2. Effect of wheel width

The effects of wheel width are analyzed by comparing the measured data on the wheels Wh2 ($r = 135$ mm, $b = 110$ mm) and Wh1 ($r = 135$ mm, $b = 165$ mm) in Figs. 14 and 15. The widths of the wheels are increased by 50%. Lugs with a height of 15 mm (Wh21 and W11) and 10 mm (Wh22 and W12) were installed to the wheels.

Increasing the width or radius of a wheel has a similar effect: it enhances the area of wheel–soil interaction at the same entrance angle. Hence, the width and radius have similar influence on wheel performance.

If the slip ratio is 0.6, for the wheels Wh21 and Wh11 with lugs of 15-mm height, the wheel sinkage decreases from 34.6 mm to 30.6 mm (11.6%), the drawbar pull increases from 20.8 N to 23.5 N (13.0%), and the driving torque decreases from 6.7 N m to 6.5 N m (3.0%); while for the wheels Wh22 and Wh12, the wheel sinkage decreases from 32.1 mm to 28.1 mm (12.5%), the drawbar pull increases from 19 N to 22.6 N (18.9%), and the driving torque decreases from 6.4 N m to 6.1 N m (4.7%).

Increasing the wheel width can improve the driving performance of a wheel primarily from two aspects. (1) The entrance angle of a wheel with a larger width is obviously reduced under the same slip ratio. The entrance angles of Wh1 are decreased by 5° as compared with that of Wh2. However, the slopes of the entrance angle versus the slip ratio plots are little influenced. (2) Even if the entrance angles are same, the resistance coefficient of the wheel with a larger width is smaller. RC of the wheel Wh1 is smaller than that of Wh2 by 0.04–0.05. The improvement in the drawbar pull coefficient PC is mainly benefited from a decrease in RC, rather than an increase in the tractive efficiency TC, because TC is little influenced by the wheel width.

Increasing the wheel width can improve the drawbar pull performance of a wheel by decreasing the resistance of soil, while increasing the radius of a wheel improves the wheel performance by providing more resistance moment to the motor. Increasing the wheel width is better in improving the driving performance of a wheel as compared with increasing the wheel radius. However, an increase in the wheel width might lead to an increase in the steering torque and thereby severely affect the flexibility in maneuvering the steering.

6. Influence of vertical load on wheel performance

Experimental results versus the amended slip ratio for the wheel Wh32 with different vertical loads are shown in

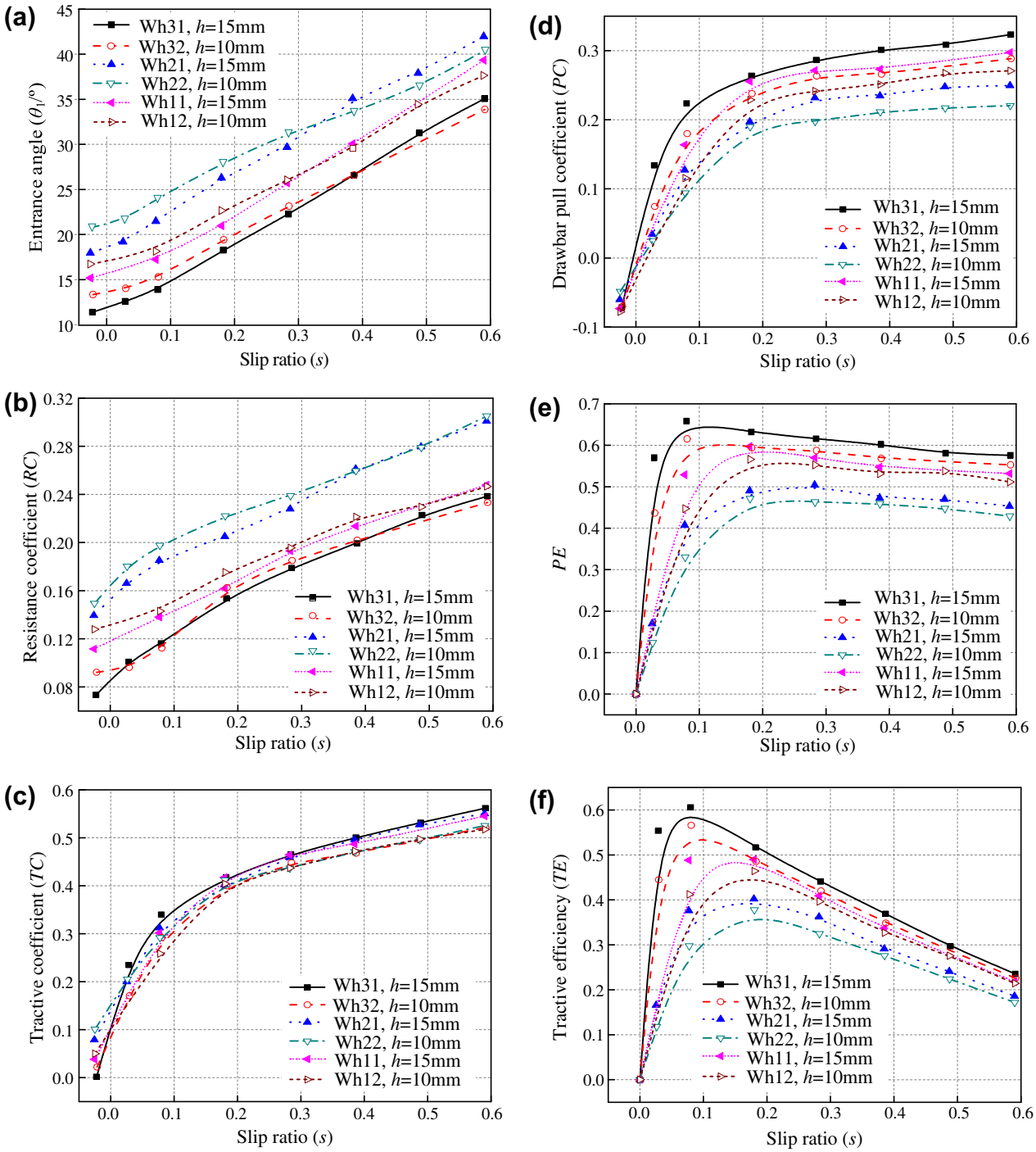


Fig. 15. Relative performance indices for wheels with different dimension.

Fig. 18. With an increase in the normal forces that are equal to the vertical loads, all the absolute performance indices of the wheel increase. If the slip ratio is 0.6 and the average normal force increases from 35 N to 80 N (150 N), the resulting wheel sinkage increases from 21.6 mm to 26.7 mm (30.4 mm), the drawbar pull increases from 14.1 N to 24.8 N (32.2 N), and the driving torque increases from 3.48 N m to 7.35 N m (11.52 N m). The relative increase in the ratio of the normal force is 128.6% (328.6%), but those in the resulting wheel sinkage, drawbar pull, and driving torque are much smaller: 23.6% (40.4%), 75.9% (128.4%), and 111.2% (231.0%), respectively. The variance of the vertical load has larger influence on the drawbar pull and resistance moment than on wheel sinkage.

The relative indices of the wheel under different vertical loads are shown in Fig. 19. The tractive coefficient TC of a wheel decreases with an increase in the vertical load, while RC increases. There are primarily two reasons that cause RC to increase: (1) If the slip ratio is same, the larger the

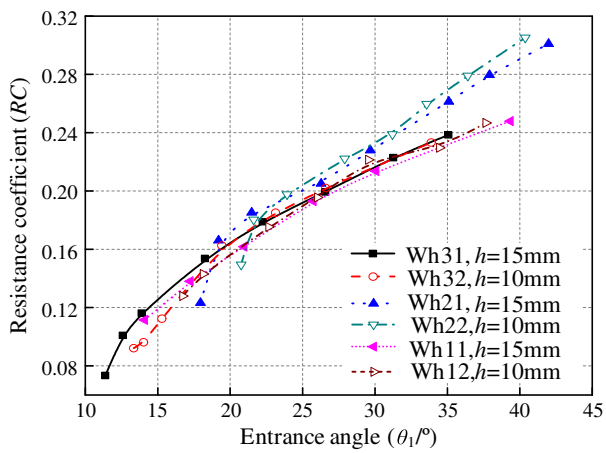


Fig. 16. Curves of resistance coefficient versus entrance angle.

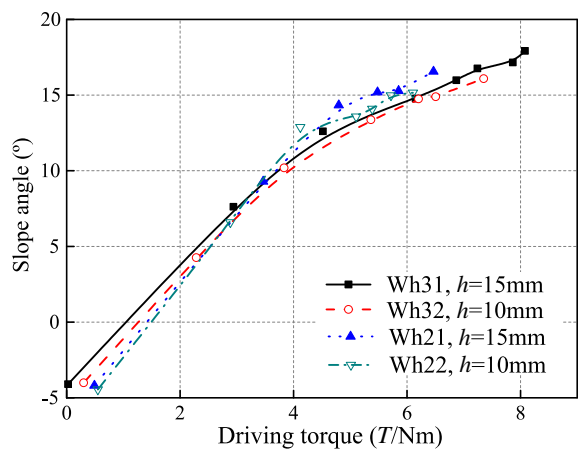


Fig. 17. Curves of slope-climbing angle versus driving torque.

vertical load, the larger the wheel sinkage and the entrance angle. Hence, *RC* increases with the vertical load, as it is an increasing function of the entrance angle. (2) Even if the entrance angle is same, a larger vertical load can cause a larger *RC*, implying that a larger compaction resistance will be generated under a larger vertical load. Both the decrease in *TC* and increase in *RC* can cause *PC* to decrease. If the slip ratio is 0.6, *PC* decreases from 0.39 to 0.29 (0.23); the corresponding slope angle that the rover could climb up decreases from 21.2° to 16.1° (12.8).

While designing a planetary rover, the influence of the vertical load, which is related to the wheel number and payload, on wheel performance should be considered to ensure the traversability and slope-climbing ability of the rover.

7. Influence of running state on wheel performance

7.1. Influence of velocity

In order to investigate the effect of velocity on planetary rovers, experiments for both smooth wheel Wh14 and lugged wheel Wh12 were performed with a vertical load

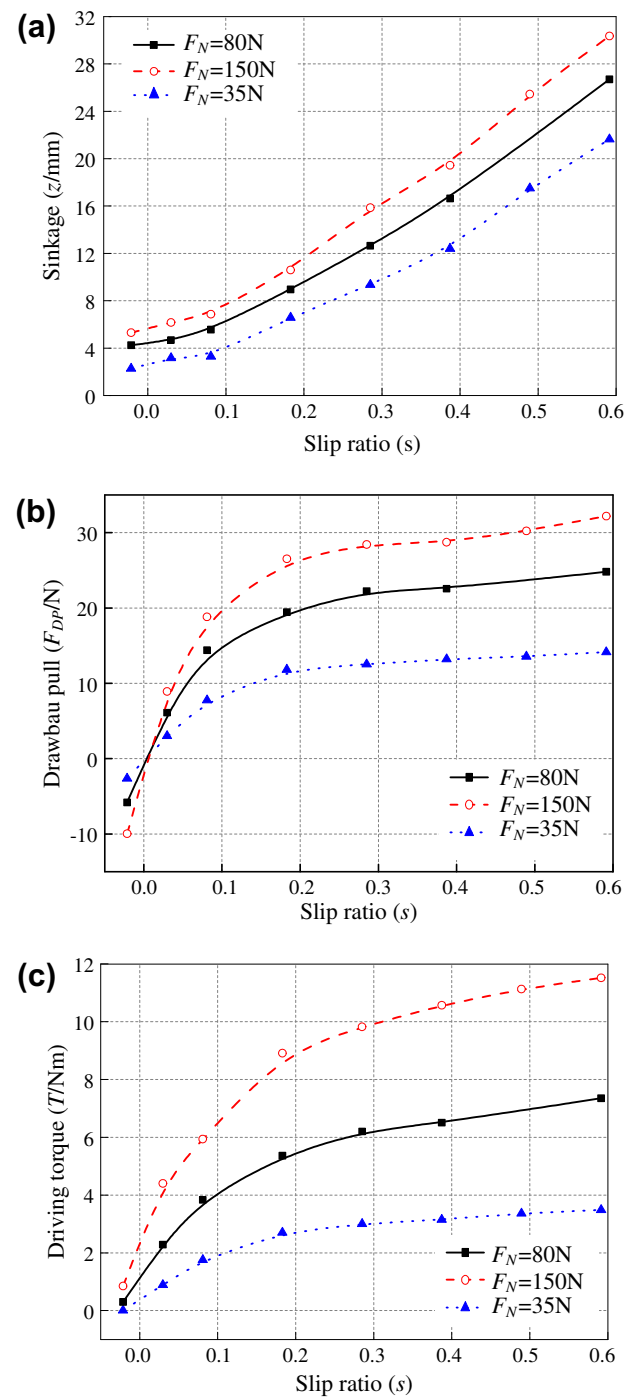


Fig. 18. z, F_{DP} and T for wheel Wh32 with different vertical loads.

of 80 N. The slip ratio was set from −0.4 to 0.4, with a step of 0.2. Fig. 20 shows the raw experimental data on lugged wheel Wh12 under the slip ratio of 0.2. The measured drawbar pull and driving torque fluctuate with different frequency, which is proportional to the velocity as calculated by Eq. (18). The velocity influences the fluctuation cycles of waves, but the mean values of the steady part are almost same. The mean sinkage, drawbar pull, and driving torque of the steady parts as well as the corresponding standard deviation (*SD*) are listed in Table 6.

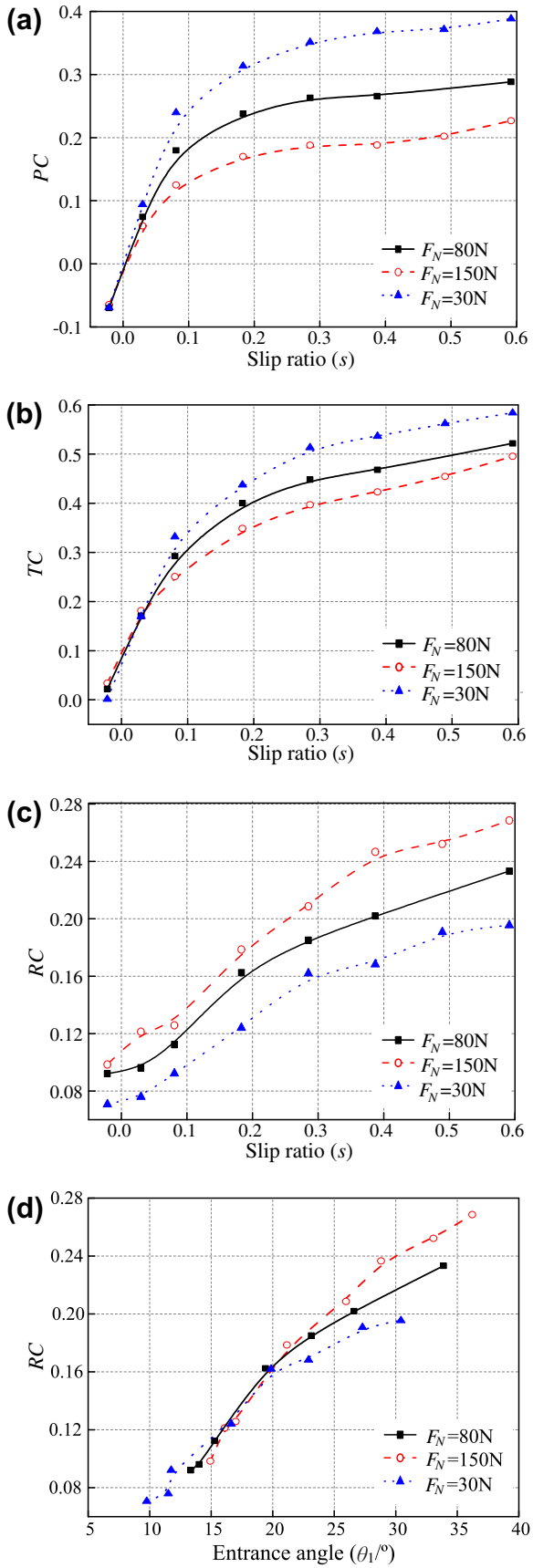


Fig. 19. Influence of normal load on relative indices of wheel Wh32.

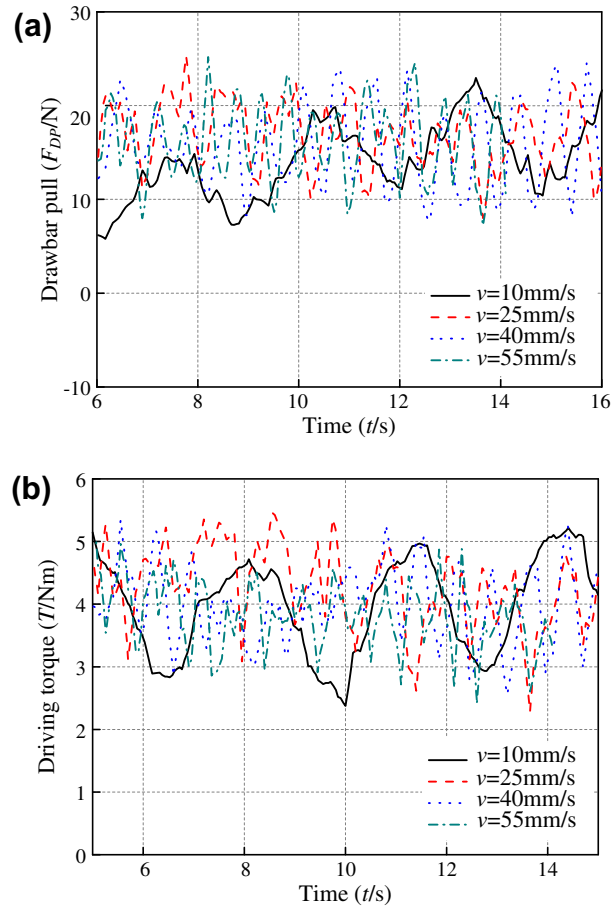


Fig. 20. Raw data of F_{DP} and T for Wheel Wh12 with different velocities ($W = 80$ N, $s = 0.2$).

By analyzing all the acquired data, it is found that the influence of velocity on the average values of sinkage, vertical load, drawbar pull, and driving torque are negligible for a low-speed planetary rover. The fluctuating cycle of measured data for a lugged wheel is proportional to the velocity, but the equivalent effect evaluated by the mean value is little influenced. Fig. 21 shows the results of drawbar pull and torque for the smooth wheel. Therefore, relatively low moving velocities (10 mm/s or 25 mm/s) were used for the experiments in order to obtain sufficient raw data within a limited moving distance.

7.2. Influence of repetitive passing

The wheel Wh12 was utilized to test the influence of repetitive passing on the wheel performance. The wheel was controlled to move four times in the same direction on the same path, as shown in Table 3. As observed from the raw data curves in Fig. 22, the raw data on F_{DP} and T are little influenced by the passing times of the wheel. The curves of TC shown in Fig. 23 indicate that the average value of the driving torque is little influenced by passing times under all the slip ratios. However, there is a difference

Table 6
Mean values and standard deviations for z, F_{DP} and T with different velocity.

Velocity (mm/s)	Mean value			*SD*		
	z (mm)	F_{DP} (N)	T (N m)	z (mm)	F_{DP} (N)	T (N m)
10	10.33	17.68	4.23	0.47	3.28	0.62
25	10.50	16.95	4.28	0.20	3.92	0.72
40	10.29	15.62	3.97	0.39	4.16	0.64
55	10.28	16.16	3.81	0.54	4.26	0.61

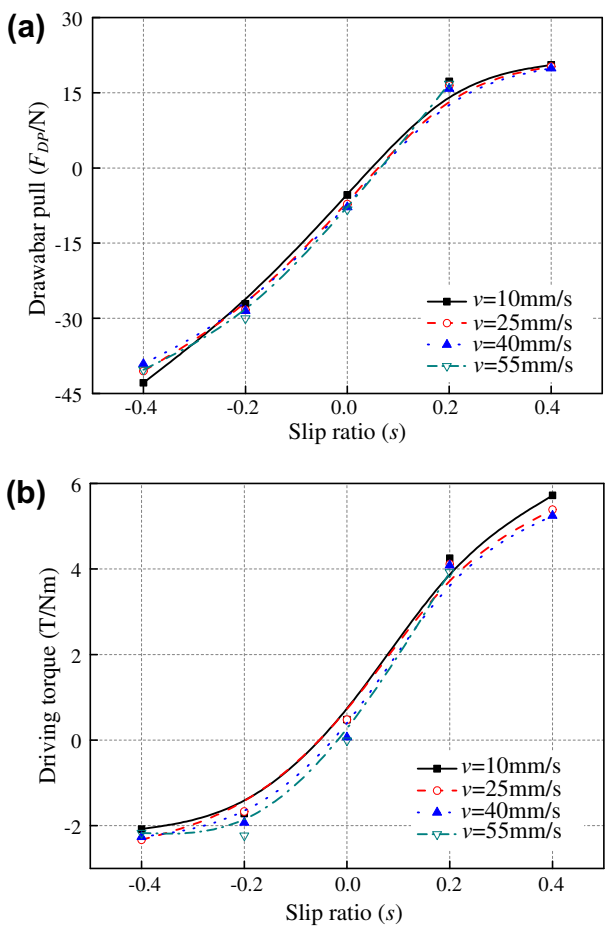


Fig. 21. Influence of moving velocity on average value of F_{DP} and T for wheel Wh14 ($W = 80$ N).

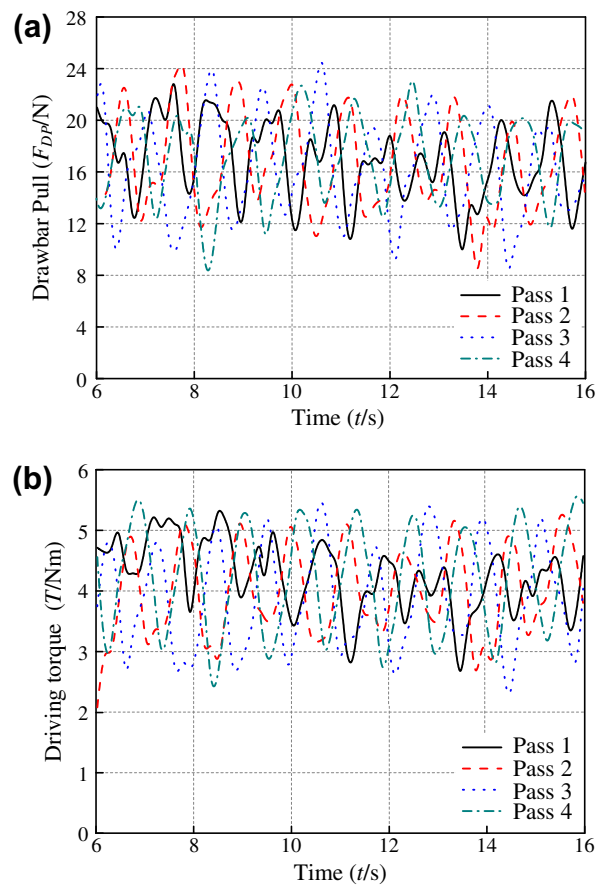


Fig. 22. Influence of repetitive passing on raw data of F_{DP} and T for wheel Wh12 ($F_N = 80$ N).

of about 1 mm between the wheel sinkage of two neighboring passings. One important reason for this is the residual deformation of soil because the wheel sinkage is measured using the undisturbed soil surface as the reference. As a conclusion, the mechanical and physical properties of soil are little influenced by the passing times of wheels. The influence of repetitive passing on wheel–soil interaction models is negligible.

7.3. Influence of entrance angle

As seen from Figs. 16 and 19d, there is an explicit relationship between the entrance angle of a wheel and the resistance coefficient of the soil. The resistance coefficient between the experimental wheel and soil is roughly a linear function of the entrance angle ranging from 0.07 to 0.31, rather than a constant value. If the entrance angle is 10°, the resistance coefficients for all the wheels are approximately 0.08; if it is 15°, the resistance coefficients are approximately 0.12. The slope of RC varying with the entrance angle is influenced by the wheel dimensions and normal force. It increases with a decrease in the wheel width or an increase in the normal force. The relationship between the entrance angle and the other performance indices are not so clear.

7.4. Influence of slip ratio

The wheel's slip ratio is an important state variable. Almost all the wheel indices are influenced by the slip ratio, necessitating their analysis under a certain slip ratio.

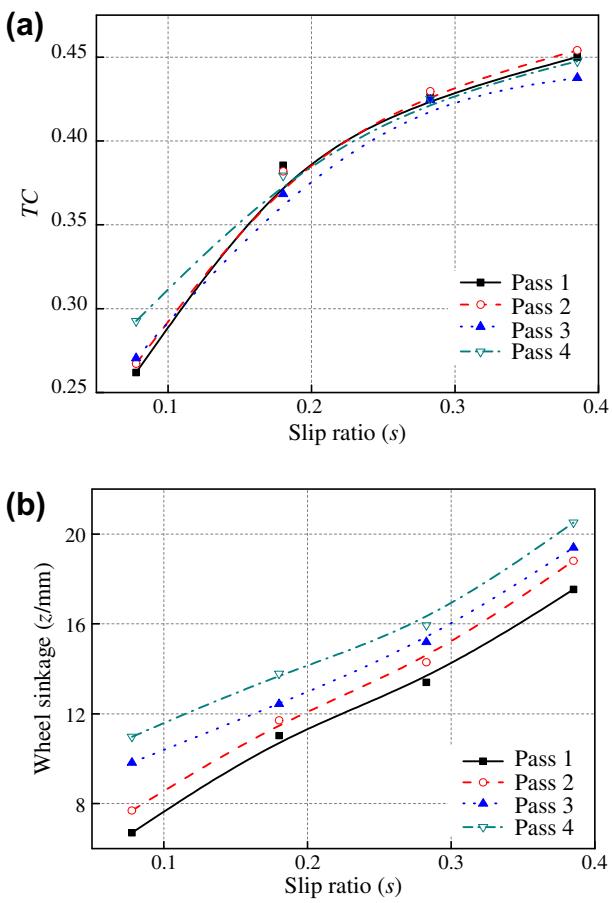


Fig. 23. Influence of repetitive passing on TC and z for Wh12 ($F_N = 80$ N).

The absolute indices such as the wheel sinkage, drawbar pull, and driving torque are increasing functions of the slip ratio. Analyzing the above figures that use the slip ratio as the independent variable, one could divide the slip ratio into three phases. (1) Rising phase ($s = 0$–0.2): during which the drawbar pull, driving torque, and PC increase rapidly. The tractive efficiency TE and the drawbar pull efficiency PE attain the maximum value in this phase. It is the ideal working phase for a rover moving on relatively flat terrain. (2) Transitional phase ($s = 0.2$–0.6): during which the drawbar pull, driving torque and PC increase slowly, PE almost keeps constant, while TE decreases rapidly. The wheels work in this phase when the rover is climbing up a slope with a large angle or negotiating obstacles. (3) High-sinkage phase ($s > 0.6$). Slip-sinkage of the wheels occurs during all the phases. However, in this phase, the wheel exhibits more severe sinkage. For example, the wheel sinkage increases from 2.89 mm to 28.53 mm if the slip ratio increases from 0 to 0.6, but the wheel sinkage reaches 52.92 mm rapidly if the slip ratio is 0.8 (Wh32, $W = 80$ N). The sinkage will increase if the slip ratio exceeds 0.8, and the wheel will get stuck by digging the soil if the slip ratio is close to 1. This is the phase that a rover should avoid. According to Eq. (15) and the curves of TC, the effective power that should be supplied by the driving motor and reduction gears increases greatly in order to keep a steady linear velocity with an increase in the slip ratio. The increased energy is consumed by the wheel slip. Therefore, a rover must move with a low linear velocity if the slip ratio is high, due to the limitation of motor power.

The wheels of a rover may have different slip ratios while moving over rough deformable terrain. It is better to coordinate the velocity of different wheels to get reasonable slip ratios for them in order to avoid high slip-sinkage and optimize the energy consumption.

8. Summary and conclusions

(1) Increasing the width and radius can improve the drawbar pull performance of a rover wheel. It is better to design the wheel with radius and width as large as possible, which is restricted by the rover dimensions. Increasing the wheel width can improve the driving performance of a wheel by decreasing the resistance of soil, while increasing the radius of a wheel improves the drawbar pull performance by providing more resistance moment to the motor. Enhancing a wheel's drawbar pull performance by increasing the wheel dimensions does not require more power from the motor, although a larger driving torque should be provided.

(2) Parameters of wheel lugs have obvious influence on wheel's driving performance. Increasing the lug height can also improve the drawbar pull performance of a wheel, which is more obvious than increasing the wheel radius. However, the ratio between the lug height and the wheel radius should be restricted because the fluctuation amplitude of wheel–soil interaction mechanics is approximately a linear function of it. There should be a sufficient number of lugs to ensure that the lug and soil interact continuously and steadily so as to achieve the maximum tractive performance. The lug inclination angle can lead to a decrease in the fluctuation amplitude of mechanics, but may cause side forces on a wheel. It is better to find the minimum angle for which the fluctuation is smallest.

(3) Decreasing the vertical load can obviously influence the wheel performance by decreasing the tractive coefficient TC and increasing the resistance coefficient RC. The influence of the vertical load on the wheel performance should be considered to ensure the traversability and slope-climbing ability of the rover.

(4) The wheel–soil interaction mechanics is little influenced by the wheel velocity and passing times of a planetary rover's wheel, which could be neglected while deducing the wheel–soil interaction models. However, the slip ratio and entrance angle are two important variables that must be considered while deducing a wheel–soil interaction model.

(5) Almost all the wheel indices are influenced by the slip ratio, which can be divided into three phases: rising phase ($s = 0$–0.2), transitional phase ($s = 0.2$–0.6), and high-sinkage phase ($s > 0.6$). While controlling a rover, it is better to coordinate the velocity of different wheels to get reasonable slip ratios for them in order to avoid high slip-sinkage and optimize the energy consumption.
(6) The resistance coefficient between an experimental wheel and soil is roughly the linear function of the entrance angle with a range of 0.07–0.31, rather than a constant value.

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